Program : **Diploma in Civil and Environmental Engineering**	
Course Code:	Course Title: **Soil Mechanics and Transportation Engineering Lab**
Semester: **4**	Credits: **1.5**
Course Category: **Program core**	
Periods per week: **3(L:0 T:0 P:3)**	Periods per semester:**45**

Course Objectives:

- To provide hands-on experience with the various laboratory experiments intended to understand the index and engineering properties of soil
- To enable the students to understand suitability of different aggregates and bitumen

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Engineering Mathematics	2
Knowledge of Construction Materials		Building Construction and Construction Materials	3
Basics of Soil Mechanics and Transportation Engineering		Soil Mechanics and Transportation Engineering	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Determine the index properties of soil	9	Applying
CO2	Determine the permeability, shear strength and compressibility of soil	14	Applying
CO3	Perform different tests on aggregates for its suitability as a road construction material	7	Applying
CO4	Perform different tests on bituminous material for its suitability for road construction	11	Applying

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Determine the index properties of soil		
M1.01	Determine water content of given soil sample by oven drying method as per IS: 2720 (Part- II).	1	Applying
M1.02	Determine specific gravity of soil by pycnometer method as per IS 2720 (Part- III).	2	Applying
M1.03	Determine Plastic limit, Liquid Limit and shrinkage limit along with Plasticity Index of given soil sample as per IS 2720 (Part- V).	3	Applying
M1.05	Determine grain size distribution of given soil sample by wet and dry sieve analysis as per IS 2720 (Part- IV).	3	Applying
CO2	Determine the permeability, shear strength and compressibility of soil		
M2.01	Determine coefficient of permeability by constant head testas per IS 2720 (Part- XVII).	1.5	Applying
M2.02	Determine coefficient of permeability by falling head test as per IS 2720 (Part- XVII).	1.5	Applying
M2.03	Determine shear strength of soil by direct shear test as per IS 2720 (Part-XIII).	2	Applying
M2.05	Determination of coefficient of consolidation by conducting consolidation test as per IS 2720-1986 (Part XV)	3	Applying
M2.06	Determination of maximum dry density and	3	Applying

	optimum moisture content by standard proctor test as per IS 2720 (Part -VII).		
	Lab Test 1	2	
CO3	Perform different tests on aggregates for its suitability as a road construction material		
M3.01	Determination of Flakiness and Elongation Index on aggregates as per IS 2386 Part IV.	2	Applying
M3.02	Determination of Angularity Number of aggregates IS 2386 Part IV	2	Applying
M3.03	Conduct Aggregate impact test as per IS 2386 Part IV.	2	Applying
M3.04	Conduct Los Angeles Abrasion test on aggregate as per IS 2386 Part IV.	2	Applying
M3.05	Conduct Aggregate crushing test as per IS 2386 Part IV.	1	Applying
CO4	Perform different tests on bituminous material for its suitability for road construction.		
M4.01	Determination of Softening point of bitumen as per IS 1205	3	Applying
M4.02	Conduct Penetration test on bitumen as per IS 1203	3	Applying
M4.03	Determination of Flash and Fire Point test of bitumen as per IS 1209.	3	Applying
M4.04	Conduct ductility test on Bitumen as per IS 1208	3	Applying
	Lab Test II	2	

Text / Reference:

T/R	Book Title/Author
1	Punmia, B.C., "*Soil Mechanics and Foundation Engineering, Laxmi Publication*", Delhi.
2	Arora, N. L., *"Transportation Engineering",* Khanna Publishers, Delhi
3	L.R. Kadiyali, *"Transportation Engineering",* Khanna Book Publishing Co., New Delhi (ISBN: 978-93-82609-858)
4	Arora ,KR, *"Soil Mechanics and Foundation Engineering",* Standard Publisher.
5	Duggal, Ajay, K. and Puri, V. P., "*Laboratory Manual in Highway Engineering*", New Age International (P) Limited, Publishers, New Delhi
6	Raj, P. Purushothama, *"Soil Mechanics and Foundation Engineering"*, Pearson India, New Delhi.
7	Khanna S.K., Justo, C E G and Veeraragavan, A., *"Highway Engineering"*, Nem Chand and Brothers, Roorkee
8	Murthy, V.N.S., "*A text book of soil mechanics and foundation Engineering",* CBS Publishers & Distributors Pvt. Ltd., New Delhi.

Online resources:

Sl.No.	Website Link
1	https://nptel.ac.in/courses/105106142
2	https://nptel.ac.in/courses/105101160
3	https://ts-nitk.vlabs.ac.in/
4	https://www.youtube.com/@nitttr/videos